

Examples of industry-specific AI use cases to tease out ideas (module 3)

Largely generated by OpenAI Deep Research using o1-pro

Industry	Predictive AI (Forecasting & Decision Optimization)	Generative AI (Content Creation & Personalization)	Agentic AI (Autonomous Actions & Decisions)
Banking & Financial	AI-driven risk models predict loan defaults and fraud, improving credit decisions and reducing losses ncino.com	Banks use genAI for customer outreach, e.g. auto-generating personalized marketing copy and communications sas.com	Autonomous trading algorithms analyze markets and execute trades in real time with minimal human input redhat.com
Insurance	Machine learning forecasts risk and flags fraud, enabling quicker underwriting and fewer claim errors spd.tech	Insurers leverage genAI to draft policies and docs automatically – e.g. AXA auto-generates policy documents tailored to customers spd.tech	AI bots settle claims end-to-end in seconds – Insurtech Lemonade's AI paid a claim in 3 seconds with no paperwork prnewswire.com
Healthcare	Predictive analytics forecast patient volumes and disease risk, optimizing staffing and care delivery openloophealth.com	Doctors use genAI assistants to summarize medical notes and patient data, saving time on documentation forbes.com	Virtual health agents engage patients and monitor treatments autonomously, providing 24/7 support and follow-up redhat.com
Pharma & Life Sciences	AI models predict drug candidate safety and trial outcomes, cutting development time and avoiding late-stage failures acc-cancer.org	Biotech firms use genAI to design new molecules for drugs, generating novel compounds and predicting their efficacy netguru.com	Autonomous lab agents can design and run experiments faster than humans – AI plans tests, analyzes data, and iterates in R&D redhat.com
Retail & E-Commerce	Retailers use AI to forecast demand and shopper trends, optimizing inventory and reducing stockouts forbes.com	GenAI creates personalized product content and ads, boosting customer engagement with AI-generated campaigns rtslabs.com	Smart store systems autonomously reorder stock and adjust shelf layouts in real time based on sales data oracle.com
Manufacturing	Predictive maintenance alerts factories to machine failures before they happen, cutting downtime forbes.com	Manufacturers employ genAI for generative design, creating optimized product designs from thousands of simulations rtslabs.com	AI-powered “lights-out” factories adjust production on the fly – agentic AI controls robots and workflow end-to-end copia.io
Supply Chain & Logistics	AI forecasts supply-chain demand (e.g. PepsiCo improved forecast accuracy by 10%), avoiding shortages and delays rtslabs.com	Logistics providers use genAI to generate shipment reports and customer updates automatically, speeding up communications dhl.com	Autonomous supply-chain agents manage routing and inventory – AI handles warehouse picking and fleet scheduling without human intervention redhat.com
Automotive & Mobility	Automakers use AI to predict vehicle maintenance needs – Volvo's platform cut unplanned truck stops by 25% cognitgo.ai	GenAI simulates driving scenarios (weather, traffic) to train autonomous vehicle systems for safer performance forbes.com	Self-driving vehicles operate with agentic AI – Waymo runs a 24/7 ride-hailing service with no safety drivers on board capitalone.com
Energy & Utilities	Utilities apply AI to forecast energy demand and detect grid issues early – e.g. monitoring turbine vibrations to predict faults smartdev.com	GenAI chatbots handle customer service for utilities, automating routine inquiries and outage reports 24/7 pwc.com	Autonomous grid control is emerging – AI agents balance power supply and demand and isolate faults in real time on the grid blog.siemens.com
Agriculture	Farmers use AI to predict crop yields and pests, enabling timely interventions and boosting output farmcrediteast.com	GenAI-powered virtual agronomists analyze weather/soil data and advise farmers on crop strategies via natural language mckinsey.com	Autonomous farm equipment operates with minimal oversight – e.g. John Deere's AI-driven tractor plows fields with GPS and sensors, no driver needed ces.tech
Education	Schools deploy AI to identify at-risk students early, allowing advisors to intervene and improve graduation rates aera.net	Educators use genAI tutors (e.g. Khan Academy's GPT-4 tutor) to personalize learning, providing on-demand explanations and feedback openai.com	AI assistants automate grading and admin tasks, allowing teachers to focus on students – AI handles routine work in the background weforum.org
Government & Public	Agencies use predictive tech to prevent problems – from anticipating power outages to spotting public health outbreaks before they spread statescoop.com	Governments employ genAI to draft documents and reports – AI assistants can compose emails, summaries and even first drafts of policies for review gsa.gov	Smart city systems act autonomously – e.g. AI-managed traffic lights in Seattle adjust signal timing in real time to reduce congestion without human intervention planning.org
Legal Services	Law firms leverage AI to predict case outcomes – analyzing past cases to assess win probabilities and inform strategy (e.g. 80% win chance prediction) americanbar.org	Lawyers use genAI to draft and review contracts faster – AI systems summarize large contracts and flag key clauses or risks in minutes attorneyatwork.com	Agentic AI legal assistants are on the horizon – autonomous AI agents can be tasked with legal research or drafting, self-check their work, then hand off to attorneys for final review thomsonreuters.com
Real Estate & Construction	Realtors use AI to predict property values (e.g. Zillow's AI-driven Zestimate for home pricing) for smarter pricing and investment decisions zillow.mediaroom.com	GenAI tools auto-generate property listings and ads – agents input home features and AI writes a compelling listing description in seconds gsascendix.com	Autonomous construction machines are being deployed – Built Robotics retrofits excavators to dig and operate on sites with AI guidance and no human driver freethink.com
Marketing & Advertising	Marketers use AI to predict campaign results – analyzing past data to forecast customer responses and conversion rates, improving ad spend ROI improvado.io	Generative AI creates ad content at scale – e.g. Coca-Cola's 2023 campaign used AI-generated imagery and video for its holiday ads, blending human creativity with AI speed emarketer.com	Autonomous marketing platforms (e.g. Albert.ai) run campaigns across channels, dynamically optimizing ads and budget with minimal human oversight brainpod.ai
Media & Entertainment	Streaming services use AI to predict hit content – by analyzing viewer trends, they greenlight shows likely to succeed and recommend what each user will watch next forbes.com	Studios and artists use genAI to produce content – e.g. an AI-generated song imitating Drake & The Weeknd went viral, demonstrating AI's ability to create music that fans enjoy npr.org	AI-driven game NPCs and characters can act autonomously – new games are integrating generative AI NPCs that carry on unscripted dialogues and adapt to players, creating dynamic storylines inworld.ai
Telecommunications	Telcos apply AI for network and customer predictions – ML flags network failures and predicts customer churn so providers can pre-empt outages or retain clients medium.com	Generative AI enhances telco customer support – AI chatbots handle common service requests and troubleshoot issues in natural language, improving response speed and consistency subex.com	Self-optimizing networks are becoming reality – e.g. Nokia's AI-powered SON autonomously executed 10,000+ network tuning actions, optimizing a carrier's mobile network with little human input telecomreview.com
Cybersecurity	Security teams use AI to detect threats proactively – for example, Microsoft's AI scans 78 trillion signals daily to spot cyber attacks earlier and fortify defenses news.microsoft.com	Analysts deploy genAI copilots to investigate incidents – e.g. GPT-4-based Security Copilot summarizes attack data and suggests remediation steps, speeding up incident response theverge.com	Autonomous cyber defense is emerging – agentic AI monitors networks and can isolate breaches or block malware in real time without waiting for human approval redhat.com
Defense & Aerospace	Militaries use AI for predictive maintenance – systems analyze fighter jet sensor data to predict part failures before missions, improving aircraft readiness lockheedmartin.com	Defense is tapping genAI for training and simulation – smarter war-game AIs act as realistic adversaries, forcing trainees to face unscripted tactics and learn in dynamic scenarios nextgov.com	Autonomous drones & robots are deploying – swarms of AI-powered drones can coordinate attacks or surveillance on their own, a capability poised to reshape military operations axios.com
Mining & Metals	Mining companies use AI to predict equipment health – enhanced predictive systems monitor mining processes in real time to detect hazards and prevent breakdowns mining-technology.com	Firms leverage genAI in mining for analysis and reporting – e.g. Rio Tinto uses generative AI to summarize geoscience research findings, speeding up knowledge sharing mining-technology.com	Autonomous mining vehicles boost productivity – Rio Tinto operates over 130 driverless haul trucks in its iron ore mines, running 24/7 under AI control to move materials efficiently riotinto.com